

7.11 PreQuiz: Logarithmic and Exponential Functions (calculator)

1. [Maximum mark: 6]

Lina invests \$2400 in an account that pays interest of 6% per annum, compounded quarterly. No further deposits or withdrawals are made.

- (a) Write down the quarterly growth factor for the value of the account. [1]
- (b) Find the value of the account at the end of 5 years, correct to the nearest cent. [2]
- (c) Find the smallest number of full years required for the value of the account to first exceed \$3400. [3]

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2. [Maximum mark: 6]

The number of members of an online study group is modelled by

$$M(t) = 5000e^{kt}, \quad k > 0$$

where t is the number of years after the group is created. After 2 years there are 7200 members.

- (a) Find the value of k . [2]
- (b) Find the number of members after 6 years, correct to the nearest whole number. [2]
- (c) Find the time, in years, when the number of members first reaches 10 000. Give your answer correct to two decimal places. [2]

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